

**LAPORAN TUGAS PRAKTIK KOMPREHENSIF -  
STRUKTUR SISTEM OPERASI**



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# Tugas Praktik Komprehensif

## Tugas 1: Process Manager

```
putrafernanda@DESKTOP-TT1 x + v
GNU nano 7.2 process_manager.c

#define _XOPEN_SOURCE 700
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/time.h> // gettimeofday
#include <sys/wait.h> // waitpid, macros
#include <string.h>
#include <sys/types.h>
#include <sys/ipc.h>
#include <sys/shm.h> // shmget, shmat
#include <signal.h>
#include <errno.h>

#define CHILD_COUNT 3
#define SHM_SIZE 4096

pid_t child_pids[CHILD_COUNT];
int pipes_arr[CHILD_COUNT][2];
int shmid = -1;
int use_shm_for_child = -1; // which child uses shared memory (-1 = none)
volatile sig_atomic_t sigint_received = 0;
int nim_last_digit = 0;

void sigint_handler(int signo) {
    // Signal handler for SIGINT in parent (to gracefully terminate children).
    sigint_received = 1;
    // We DO NOT use non-async-signal-safe calls here except kill (which is safe).
    for (int i = 0; i < CHILD_COUNT; ++i) {
        if (child_pids[i] > 0) {
            kill(child_pids[i], SIGTERM); // request child termination
        }
    }
}

int main() {
    // ... (rest of the code) ...
}
```

```
putrafernanda@DESKTOP-TT1 x + v
blleh
^C

putrafernanda@DESKTOP-TT1:~$ nano process_manager.c
putrafernanda@DESKTOP-TT1:~$ gcc -Wall -O2 -o process_manager process_manager.c
putrafernanda@DESKTOP-TT1:~$ ./process_manager
=== Simple Process Manager (fork, pipe, shm, waitpid) ===
Masukkan digit terakhir NIM Anda (0-9): 9
NIM akhir GANJIL -> Child 2 akan menggunakan shared memory untuk komunikasi.
Input untuk Child 1 (faktorial): masukkan angka non-negatif (mis. 5): 4
Input untuk Child 2 (prima): masukkan batas (mis. 30): 17
Input untuk Child 3 (reverse string): masukkan string (max 1000 chars): 500
Parent: created Child 1 with PID 968
Parent: created Child 2 with PID 969
Parent: created Child 3 with PID 970
Parent: Child index 1 finished. PID=968. Exited normally, exit status=0. Execution time (approx): 0.685 ms
Result from Child 1 via pipe:
Child 1 (PID 968): faktorial(4) = 24
Parent: Child index 2 finished. PID=969. Exited normally, exit status=0. Execution time (approx): 0.635 ms
Result from Child 2 via shared memory:
2 3 5 7 11 13 17
Parent: Child index 3 finished. PID=970. Exited normally, exit status=0. Execution time (approx): 0.433 ms
Result from Child 3 via pipe:
Child 3 (PID 970): reverse('500') = '005'
Parent: shared memory segment removed.
Parent: all children handled. Exiting.
putrafernanda@DESKTOP-TT1:~$ nano file_monitor.c
putrafernanda@DESKTOP-TT1:~$ gcc -Wall -O2 -o file_monitor file_monitor.c
file_monitor.c: In function 'main':
file_monitor.c:65:5: warning: ignoring return value of 'scanf' declared with attribute 'warn_unused_result' [-Wunused-result]
    65 |         scanf("%d", &nim_last_digit);
        |         ^~~~~~
putrafernanda@DESKTOP-TT1:~$ ./file_monitor /path/direktori
Masukkan digit terakhir NIM Anda (0-9): 9
NIM ganjil -> mencetak notifikasi email dummy
inotify_add_watch: No such file or directory
putrafernanda@DESKTOP-TT1:~$ ./file_monitor /path/direktori
Masukkan digit terakhir NIM Anda (0-9): 39
NIM ganjil -> mencetak notifikasi email dummy
inotify_add_watch: No such file or directory
putrafernanda@DESKTOP-TT1:~$ nano myshell.c
```

## Tugas 2: File Monitor

```
putrafernanda@DESKTOP-TTI x + v
GNU nano 7.2 file_monitor.c *
/*
 * file_monitor.c (versi manual)
 *
 * Program memantau direktori /tmp/monitor_dir
 * dan mencatat event IN_CREATE, IN_MODIFY, IN_DELETE.
 *
 * Tidak butuh argumen di command line.
 * Akan minta digit terakhir NIM lewat input.
 */

#define _GNU_SOURCE
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/inotify.h>
#include <sys/types.h>
#include <sys/stat.h>
#include <fcntl.h>
#include <signal.h>
#include <string.h>
#include <time.h>
#include <errno.h>
#include <stdint.h>
#include <limits.h>

#ifndef NAME_MAX
#define NAME_MAX 255
#endif

#define BUF_LEN (10 * (sizeof(struct inotify_event) + NAME_MAX + 1))

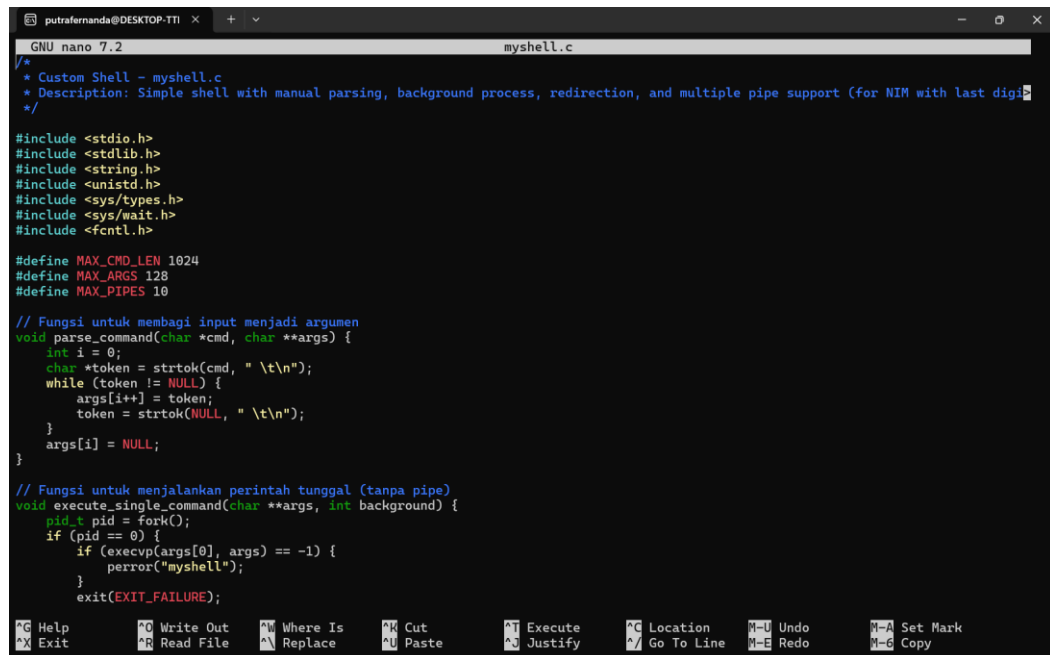
static volatile sig_atomic_t running = 1;
static int inotify_fd = -1;
static int watch_fd = -1;
static int log_fd = -1;
static int nim_last_digit = 0;

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^J Execute    ^C Location   ^U Undo       ^-A Set Mark
^X Exit      ^R Read File  ^N Replace    ^U Paste      ^_ Justify    ^_ Go To Line ^E Redo       ^-G Copy

putrafernanda@DESKTOP-TTI x + v
putrafernanda@DESKTOP-TTI:~$ nano file_monitor.c
putrafernanda@DESKTOP-TTI:~$ gcc -Wall -O2 -o file_monitor file_monitor.c
putrafernanda@DESKTOP-TTI:~$ ./file_monitor
=== File Monitor Sistem Operasi ===
Direktori default: /tmp/monitor_dir
Log file: /tmp/monitor.log
Masukkan digit terakhir NIM Anda: 9

Monitoring dimulai... (Ctrl+C untuk berhenti)
```

## Tugas 3: Custom Shell



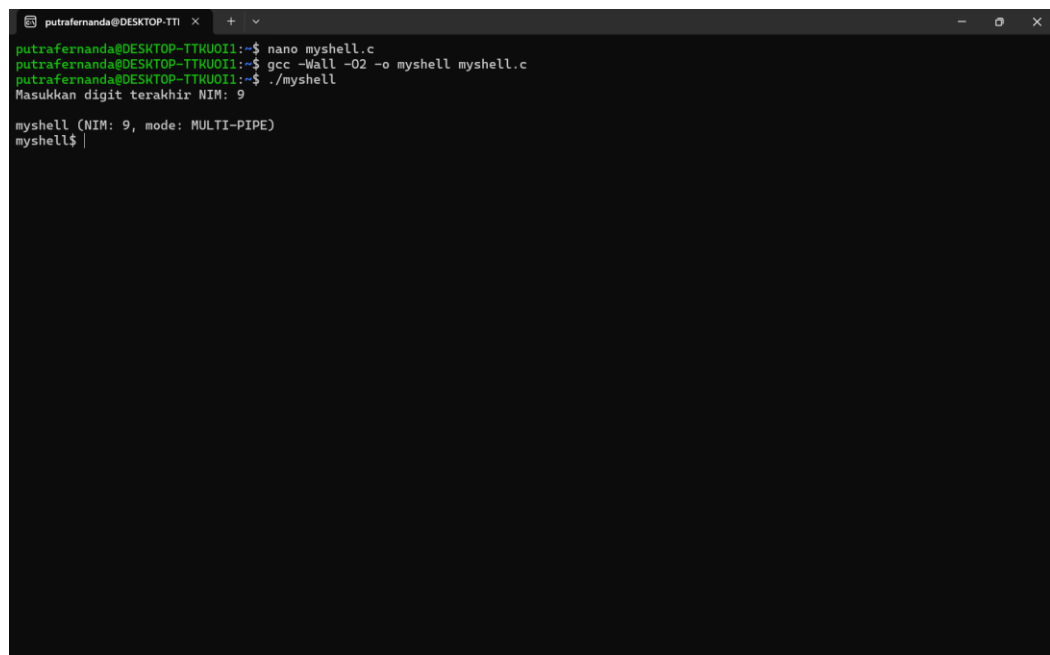
```
GNU nano 7.2 myshell.c
/*
 * Custom Shell - myshell.c
 * Description: Simple shell with manual parsing, background process, redirection, and multiple pipe support (for NIM with last digi
 */

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <fcntl.h>

#define MAX_CMD_LEN 1024
#define MAX_ARGS 128
#define MAX_PIPES 10

// Fungsi untuk membagi input menjadi argumen
void parse_command(char *cmd, char **args) {
    int i = 0;
    char *token = strtok(cmd, " \t\n");
    while (token != NULL) {
        args[i++] = token;
        token = strtok(NULL, " \t\n");
    }
    args[i] = NULL;
}

// Fungsi untuk menjalankan perintah tunggal (tanpa pipe)
void execute_single_command(char **args, int background) {
    pid_t pid = fork();
    if (pid == 0) {
        if (execvp(args[0], args) == -1) {
            perror("myshell");
        }
        exit(EXIT_FAILURE);
    }
}
```



```
putrafernanda@DESKTOP-TTKU011:~$ nano myshell.c
putrafernanda@DESKTOP-TTKU011:~$ gcc -Wall -O2 -o myshell myshell.c
putrafernanda@DESKTOP-TTKU011:~$ ./myshell
Masukkan digit terakhir NIM: 9

myshell (NIM: 9, mode: MULTI-PIPE)
myshell$
```

## Tugas 4: System Information Tool

```
putrafernanda@DESKTOP-TT1: ~$ nano sysinfo_tool.c
GNU nano 7.2 sysinfo_tool.c
/*
 * System Information Tool
 * -----
 * - Membaca file /proc/ menggunakan open() dan read()
 * - Menampilkan informasi CPU, Memory, Disk, Process, dan Network
 * - Tambahan: CPU load average atau TCP socket info tergantung NIM terakhir
 */

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <dirent.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/statvfs.h>
#include <sys/types.h>
#include <ctype.h>
#include <errno.h>

#define BUF_SIZE 8192
#define PATH_MAX_LEN 512

int nim_last_digit;

/* Fungsi pembantu membaca isi file /proc menjadi string */
void read_file(const char *path, char *buffer, size_t size) {
    int fd = open(path, O_RDONLY);
    if (fd < 0) {
        snprintf(buffer, size, "N/A");
        return;
    }
    ssize_t bytes = read(fd, buffer, size - 1);
    if (bytes > 0) buffer[bytes] = '\0';
    else buffer[0] = '\0';
    close(fd);
}

[ Read 228 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        [ Read 228 lines ]
^X Exit      ^R Read File  ^N Replace    ^U Paste      ^J Execute
^_ Location  ^U Undo       ^M-A Set Mark
^_ Go To Line ^E Redo       ^M-G Copy
```

```
putrafernanda@DESKTOP-TT1: ~$ nano sysinfo_tool.c
putrafernanda@DESKTOP-TT1: ~$ gcc -Wall -O2 sysinfo_tool.c -o sysinfo_tool
putrafernanda@DESKTOP-TT1: ~$ ./sysinfo_tool
Masukkan digit terakhir NIM: 9

=== SYSTEM INFORMATION TOOL ===

----- CPU INFO -----
| Model: AMD Ryzen 5 6600H with Radeon Graphics |
| Cores: 6 | Frequency: 3293.81 MHz |
-----
----- MEMORY INFO -----
| Total: 6966480 kB | Used: 446280 kB | Free: 6334132 kB |
-----
----- DISK INFO -----
| Total: 1031018 MB | Used: 2281 MB | Free: 1028737 MB |
-----
----- PROCESS LIST -----
| PID | Process Name | Status |
-----
| 1 | systemd | S |
| 2 | init-systemd(Ub | S |
| 7 | init | S |
| 54 | systemd-journal | S |
| 104 | systemd-udev | S |
| 130 | systemd-resolve | S |
| 148 | systemd-timesyn | S |
| 180 | cron | S |
| 181 | dbus-daemon | S |
| 194 | systemd-logind | S |
| 196 | wsl-pro-service | S |
| 206 | sshd | S |
| 214 | rsyslogd | S |
| 218 | agetty | S |
| 221 | agetty | S |
| 222 | unattended-upgr | S |
| 326 | login | S |
| 371 | systemd | S |
| 372 | (sd-pam) | S |
| 391 | bash | S |
```